

System Architecture & Design

Book a Doctor Healthcare Platform • Folder 3: System Design

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Submission Date:	July 2026

Folder 3 Metadata & Registry

Registry Parameter	Value Specification
Project Stage	Folder 3 – System Design & Architecture
Primary Objective	MERN Stack Mapping, Database Schema, and Interface Layouts
Status	Design Verified - Architecture Approved

1. System Architecture Overview

The "Book a Doctor" application utilizes a modular MERN architecture. This design pattern ensures clear separation of concerns, scalability for growing patient datasets, and high-performance communication between the client-side UI and backend API routes.

2. Logical Data Schema

The MongoDB schema structure organizes healthcare information into discrete relational collections:

Collection	Stored Attributes
Users	User_ID, Name, Role (Doctor/Patient), Auth_Credentials
Doctors	Doctor_ID, Specialization, Availability_Schedule, Clinic_Location
Appointments	Booking_ID, User_ID, Doctor_ID, Status, Timestamp

3. User Interface Design & Navigation

The design workflow provides optimized paths for all three platform stakeholders:

- **Patient Workflow:** Search providers -> Review profiles -> Book slot -> Secure document transfer.

- **Doctor Workflow:** Review dashboard -> Manage schedule -> View pending consultation alerts.
- **Admin Workflow:** Maintain platform security, user roles, and system status monitoring.

4. Folder 3 Design Conclusion

The System Design phase is complete. The architectural foundation, database schema models, and user interface workflows have been verified and approved, enabling the development team to proceed with implementation.

Prepared By:

Team Lead Approval:

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